SET-B

Punyashlok Ahilyadevi Holkar Solapur University, Solapur

B.Sc(ECS) Part-II Sem-III (CBCS) (NEP w.e.f. June 2025) Practical Examination, Oct-2025

Practical VSC1 Based on DSC1-3 (Data Structures using C++) and DSC1-4 (DBMS)

Day and Date:- Session: III

Time: -2 Hr Marks: - 30

Instructions:

1. Attempt any one questions from each section.
2. Each question carries 10 marks.
3. Journal submission: 05 marks
4. Viva(oral examination): 05 marks

Section-I (Data Structure Using C++)

1. Write a program to create and display a doubly linked list.

2. Write a program to perform preorder, inorder, and postorder traversal of a binary tree.

Section-II (DBMS)

Create the following tables with necessary constraints and insert records in each table.

Tour Table

tour_id	tour_name	destination	cost (₹)
201	Royal Rajasthan	Jaipur	25000
202	Kerala Backwaters	Kochi	18000
203	Himalayan Retreat	Manali	35000
204	Desert Safari	Jaisalmer	15000
205	Temple Heritage	Madurai	12000

Guide Table

guide_id	guide_name	experience	tour_id
1	Mohan Sharma	Expert	201
2	Meena Patel	Intermediate	202
3	Manish Verma	Beginner	203
4	Riya Das	Expert	204
5	Mahesh Rao	Intermediate	205

Q1. Solve the following Queries:

a. Display tours that have a cost above the average cost of all tours.
b. Display guide details along with their corresponding tour names.
c. Display all tours, including those without guides.
d. Display guides assigned to tours with cost above the average cost.
e. Display tours that have at least one guide.

Q2. Solve the following Queries:

a. Create a sequence named tour_seq starting from any number, increment by 2.
b. Create a view named PREMIUM_TOURS showing tours with cost greater than 30000.